Project Title

Plagiarism Detection For Academic Submissions

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
GOUTHAM VIKAS PARIMI - 2520090096
Team Number: 23

Section: 11

Under the guidance of

Anitha Chekuri

Assistant Professor, CS & IT

Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "Plagiarism Detection Engine For Academic Submissions" is a record of Bonafide work of GOUTHAM VIKAS PARIMI - 2520090096 submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

GOUTHAM VIKAS PARIMI – 2520090096

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled “Plagiarism Detection Engine For Academic Submissions” is a bonafide work done and submitted by GOUTHAM VIKAS PARIMI - 2520090096 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, <<Guide Name>>, <<Designation>>, <Department>> which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

GOUTHAM VIKAS PARIMI - 2520090096

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	
2.	Introduction	
3.	Literature survey	
4.	Problem Statement and Objectives	
5.	Hardware and Software requirements	
6.	Implementation	
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1.ABSTRACT

Plagiarism Detection Engine for Academic Submissions is an intelligent web-based system designed to identify similarities between student assignments, reports, and research documents. The system allows users to upload documents in various formats and automatically analyzes the content using Natural Language Processing (NLP) and text similarity algorithms such as TF-IDF and Cosine Similarity. It generates a detailed plagiarism report highlighting matched content, similarity percentages, and possible sources. The platform also provides an administrator dashboard to manage users, submissions, and reports securely. By promoting academic integrity, reducing manual verification efforts, and ensuring originality, the system helps educational institutions streamline the evaluation process. The project emphasizes accuracy, efficiency, scalability, and user-friendly interaction while maintaining data privacy and secure document handling. It can be further enhanced with AI-based semantic similarity detection and integration with institutional learning management systems (LMS).

By : GOUTHAM VIKAS PARIMI - 2520090096

2. INTRODUCTION

The Plagiarism Detection Engine for Academic Submissions is a web-based application designed to detect similarities and identify potentially plagiarized content in academic documents such as student assignments, project reports, essays, and research papers. Academic integrity is an essential part of education, but manually checking large numbers of submissions for copied content is time-consuming and difficult. Therefore, an automated plagiarism detection system can help students, faculty members, and educational institutions evaluate the originality of academic work more efficiently.

The proposed system allows users to upload documents in various formats and analyzes their textual content using Natural Language Processing (NLP) techniques and text similarity algorithms such as Term Frequency–Inverse Document Frequency (TF-IDF) and Cosine Similarity. It compares the submitted documents with available reference documents and identifies matching content, similarity percentages, and possible sources. The system generates a detailed plagiarism report that helps users understand the extent of similarity and make necessary corrections before submitting their work.

The application also includes an administrator dashboard to manage users, document submissions, and generated reports securely. It focuses on improving accuracy, reducing manual effort, protecting document privacy, and providing a simple, user-friendly interface. In the future, the system can be enhanced by incorporating AI-based semantic similarity detection and integration with institutional Learning Management Systems (LMS). This project aims to promote academic honesty, encourage original writing, and simplify the evaluation process in educational institutions.

3. LITERATURE SURVEY

Existing plagiarism detection systems such as Turnitin, Grammarly, Quetext, Copyscape, Scribbr, and ithenticate help identify copied or similar content in academic documents and online sources. These systems compare text, highlight matching content, and generate reports to help students and teachers check originality. Some tools also provide writing suggestions and citation support. However, the proposed Plagiarism Detection Engine for Academic Submissions aims to provide a simple and user-friendly platform for checking academic documents using TF-IDF and Cosine Similarity. It will generate similarity reports and include an administrator dashboard to manage users and submissions, helping promote academic honesty and reduce manual checking efforts.

4. Problem Statement and Objectives

The Plagiarism Detection Engine for Academic Submissions aims to identify copied and similar content in assignments, project reports, and research papers. Manual plagiarism checking takes time and effort, so this project provides a simple web-based application where users can upload documents and check their originality. It uses TF-IDF and Cosine Similarity to compare text, calculate similarity scores, and generate reports. The main objective is to reduce manual work, save time, and promote academic honesty by encouraging students to submit original work.

5. HARDWARE AND SOFTWARE REQUIREMENTS

The Plagiarism Detection Engine for Academic Submissions requires a computer or laptop with at least 4 GB RAM, a dual-core processor, and sufficient storage space to develop and run the application. The software requirements include Windows or Linux operating system, Python programming language, Flask framework, HTML, CSS, JavaScript, and a database such as SQLite or MySQL. Libraries such as Scikit-learn and NLP tools are used for text processing and similarity checking. A web browser and a code editor such as Visual Studio Code are also required for development and testing.

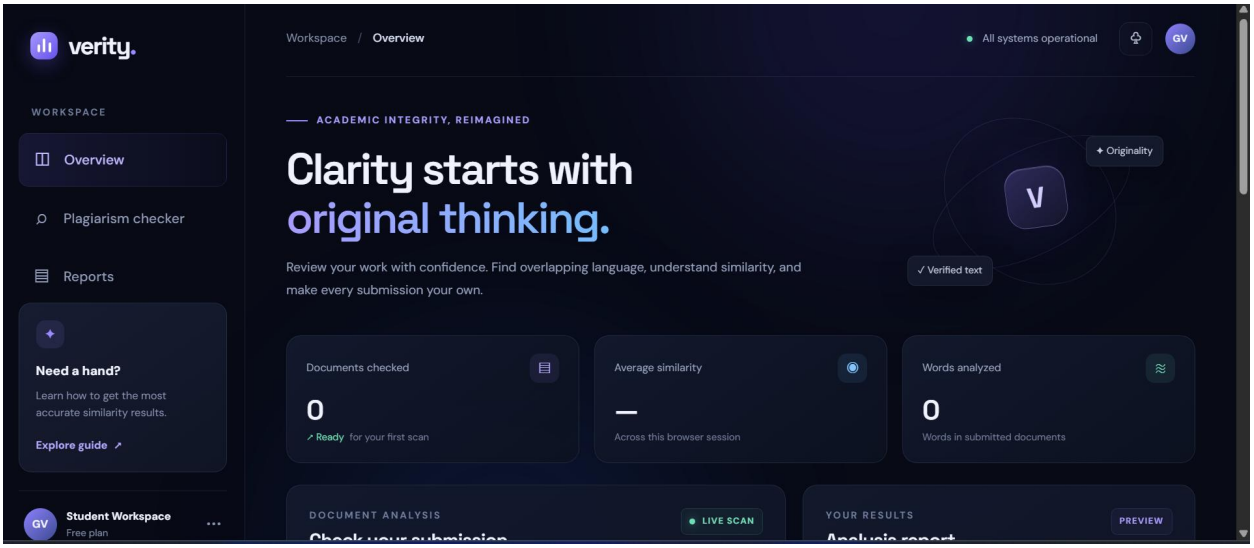
6. IMPLEMENTATION

```
1
2   <!DOCTYPE html>
3   <html lang="en">
4   <head>
5     <meta charset="UTF-8">
6     <meta name="viewport" content="width=device-width, initial-scale=1.0">
7     <title>Verity | Plagiarism Detection Engine</title>
8     <link rel="stylesheet" href="style.css">
9   </head>
10  <body>
11    <div class="app">
12      <header class="topbar">
13        <a class="brand" href="#">verity<span>.</span></a>
14        <nav>
15          <a href="#dashboard">Overview</a>
16          <a href="#checker">Plagiarism Checker</a>
17          <a href="#report">Reports</a>
18        </nav>
19        <span class="badge">BETA VERSION</span>
20      </header>
21
22      <main id="dashboard">
```

```
1
2   @import url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500
3
4   :root {
5     --bg: #080b16;
6     --panel: #111729;
7     --border: rgba(159, 174, 220, 0.15);
8     --text: #f1f4ff;
9     --muted: #8993b0;
10    --purple: #9b8cff;
11    --purple-dark: #7061ef;
12    font-family: 'DM Sans', sans-serif;
13  }
14
15  * {
16    box-sizing: border-box;
17  }
18
19  html {
20    scroll-behavior: smooth;
21  }
22
```

```
1
2   const $ = (selector) => document.querySelector(selector);
3
4   const form = $("#scanForm");
5   const documentText = $("#documentText");
6   const referenceText = $("#referenceText");
7   const fileInput = $("#fileInput");
8   const wordCount = $("#wordCount");
9   const toast = $("#toast");
10
11  let latestReport = null;
12  let checkedCount = 0;
13  let totalWordsChecked = 0;
14  let similaritySum = 0;
15  let toastTimer;
16
17  const stopWords = new Set(
18    `a an the and or but if then else of at by for from in into on
19    onto to up with without is are was were be been being am do
20    does did have has had it its this that these those as not no
21    so than too very can could will would should may might must
22    we our you your they their he she his her i me my mine them
23    what which who when where why how also such about over under
```

7. RESULTS & DISCUSSION



Explanation:

Verity is a web-based application designed to help students review academic submissions and identify similarities between texts. It uses HTML for structure, CSS for styling, and JavaScript for functionality. Users can paste their documents or upload .txt files, provide reference text, and analyze overlapping vocabulary. The dashboard displays similarity percentages, matched words, highlighted text, and downloadable reports. It also provides statistics about documents checked and words analyzed.

The current version is a frontend prototype that compares user-provided text. Future improvements can include NLP techniques such as TF-IDF and cosine similarity, a Python backend, academic database integration, user authentication, and secure document storage.

8. CONCLUSION & FUTURE SCOPE

The Plagiarism Detection Engine, Verity, provides a simple and user-friendly platform for checking text similarities in academic submissions. Built using HTML, CSS, and JavaScript, it displays similarity scores, highlights matching words, and generates downloadable reports, helping students improve their work and maintain academic integrity. In the future, the project can be enhanced with NLP techniques such as TF-IDF and cosine similarity, a Python backend, academic database integration, AI-based paraphrase detection, PDF and DOCX support, user authentication, and secure document storage to provide more comprehensive and accurate plagiarism analysis.

9. REFERENCES

The development of the Plagiarism Detection Engine, Verity, is based on information from research papers, academic articles, and online documentation related to plagiarism detection, Natural Language Processing (NLP), and text similarity analysis. References include the official documentation for Python, scikit-learn, and TF-IDF and cosine similarity techniques, along with HTML, CSS, and JavaScript documentation from MDN Web Docs. These resources provide guidance for designing the user interface, processing text, comparing documents, and generating similarity reports.